

more than extensions of the frequency probability method. In fact, in many cases, subjective probabilities add value because this approach allows you to take operational issues into account rather than depend on long-run statistical regularity.

Probability Theory and the Market

Whether or not investors recognize it, virtually all the decisions they make are exercises in probability. For them to succeed, it is critical that their probability statement combines the historical record with the most recent data available. And that is Bayesian analysis in action.

“Take the probability of loss times the amount of possible loss from the probability of gain times the amount of possible gain. That is what we’re trying to do,” says Buffett. “It’s imperfect but that’s what it is all about.”¹³

A useful example to clarify the link between investing and probability theory is the practice of risk arbitrage. Pure arbitrage is nothing more than profiting from the discrepancy in the price of a security quoted in two different markets. For example, commodities and currencies are quoted in several different markets around the world. If two separate markets quote a different price on the same commodity, you could buy in one market, sell in another market, and pocket the difference.

Risk arbitrage, which is the form more commonly practiced today, involves announced corporate mergers or acquisitions. There are individuals who practice risk arbitrage on unannounced corporate events, but this is an area that Buffett avoids and so will we. In a speech to a group of Stanford University students, Buffett shared his views on risk arbitrage. “My job,” he said, “is to assess the probability of the events [announced mergers] actually transpiring and the gain/loss ratio.”¹⁴

Let’s preface Buffett’s next remarks to the Stanford students with this scenario. Suppose the Abbott Company began the day trading at \$18 per share. Then, in midmorning, it is announced that sometime this year, perhaps in six months, Abbott will be sold to the Costello Company for \$30 per share. Immediately, the share price of Abbott races to \$27, where it settles in and begins to trade back and forth.